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Subject: Artificial Intelligence

Topic: Fraud Detection System for Digital

Payments and Banking Transaction.

Submitted To: Sir Rajesh Kumar Sahab

Title: Fraud Detection System for Digital Payments and Banking Transactions

Introduction:

With the rapid growth of online banking, mobile wallets, and digital payment systems, financial fraud has become a major challenge for banks and customers. Fraudulent activities such as unauthorized transactions, identity theft, account takeovers, and credit card fraud can cause significant financial losses and damage customer trust.

A Fraud Detection System is an Artificial Intelligence (AI)-based solution designed to monitor, analyze, and detect suspicious financial transactions in real time. The system uses data analysis, machine learning algorithms, and behavioral patterns to identify unusual activities that may indicate fraud.

When suspicious behavior is detected, the system can automatically generate alerts, block transactions, request additional verification, or notify bank administrators for further investigation. This helps financial institutions enhance security, reduce financial losses, and provide a safer digital banking experience for customers.

In modern banking, fraud detection systems play a vital role in protecting users and ensuring the reliability of digital payment platforms. By continuously learning from historical transaction data, these systems become more accurate and effective in identifying new and evolving fraud techniques.

Objective of the Project

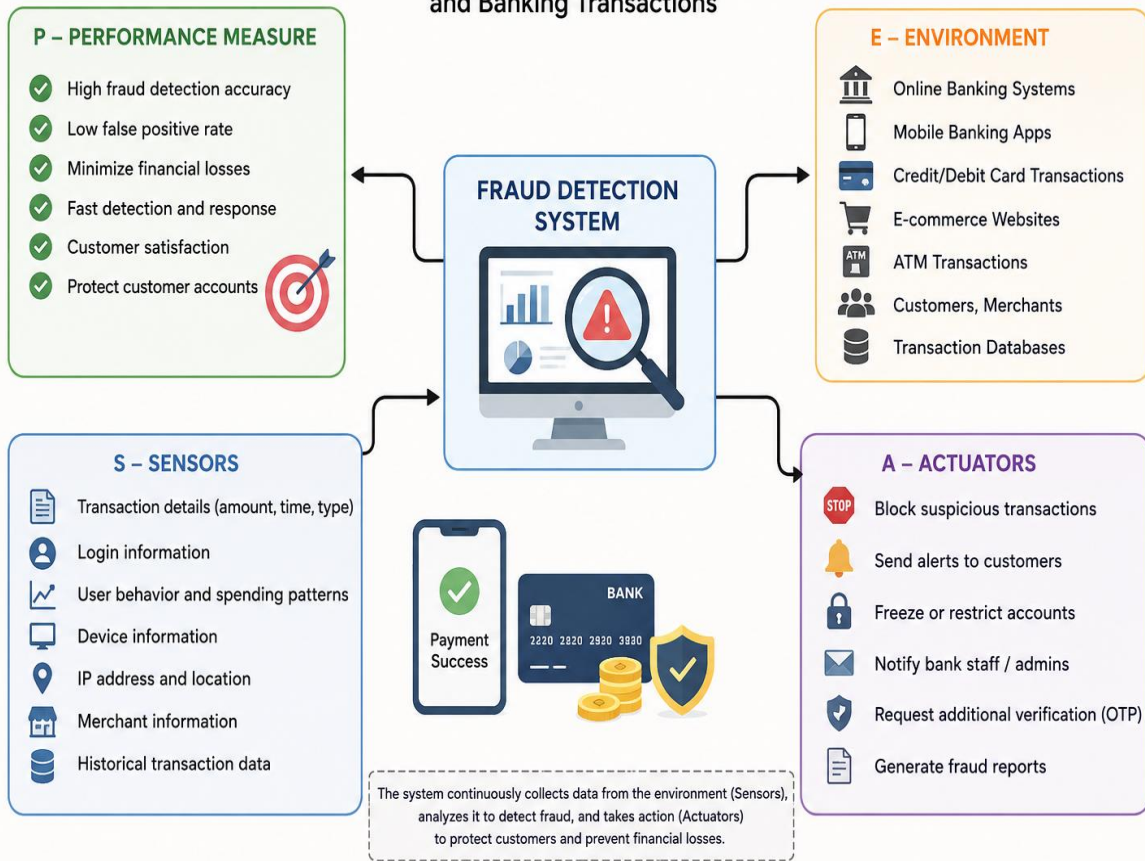
The main objective of this project is to develop an intelligent system that can:

- Detect fraudulent banking and payment transactions.
- Analyze customer transaction patterns.
- Prevent financial losses.
- Improve transaction security.
- Enhance customer trust in digital banking services.

This project demonstrates how Artificial Intelligence can be applied to solve real-world problems in the financial sector by improving security and reducing fraud risks.

PEAS Example

Fraud Detection System for Digital Payments and Banking Transactions



FRAUD DETECTION SYSTEM

FOR DIGITAL PAYMENTS AND BANKING TRANSACTIONS

AI-Powered System to Detect, Prevent and Respond to Fraudulent Transactions in Real Time

SOURCES (SENSORS)

- Transaction Details
amount, time, type
- User Behavior
spending pattern, history
- Device Information
device ID, OS, browser
- Location Data
IP address, geolocation
- Merchant Information
merchant ID, category
- Account Information
account type, balance
- Historical Data
past transactions, logs



DATA COLLECTION
Collect data from multiple sources



DATA PREPROCESSING
Clean, transform and prepare data



FRAUD DETECTION ENGINE
Machine Learning / AI Models



RISK SCORING
Assign risk score to transaction



DECISION
Fraud or Legitimate Transaction

ACTIONS (ACTUATORS)

- Block Transaction**
prevent suspicious transactions
- Send Alert**
notify customer via SMS, Email, App
- Freeze Account**
temporarily restrict account access
- Request Verification**
send OTP or additional verification
- Notify Bank Staff**
alert fraud investigation team
- Generate Report**
create fraud reports and logs

REAL-TIME MONITORING DASHBOARD

Transaction Overview

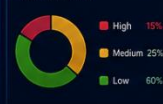


Fraud Alerts

37
High Risk Alerts

Risk Level	Count
High Risk	37
Medium Risk	62
Low Risk	153

Risk Distribution



Recent Alerts

- Txn ID: 4587 • **High Risk**
Amount: \$3,000 • Location: Unknown
- Txn ID: 4588 • **Medium Risk**
Amount: \$1,200 • New Device
- Txn ID: 4589 • **High Risk**
Amount: \$10,000 • Foreign Location

SYSTEM ARCHITECTURE



SECURITY LAYER

Encryption • Authentication • Access Control • Compliance

FEATURES

- Real-time Transaction Monitoring
- AI/ML based Fraud Detection
- Behavioral Pattern Analysis
- Adaptive Learning from New Data
- Scalable and High Performance
- Secure and Reliable System

BENEFITS

- Reduces Financial Losses
- Improves Customer Trust
- Enhances Security
- Minimizes False Positives
- Saves Time and Resources
- Supports Regulatory Compliance



PEAS Component	Description
Performance Measure (P)	Detect fraudulent transactions accurately, minimize financial losses, reduce false alarms, protect customer accounts, and provide fast detection.
Environment (E)	Online banking systems, mobile banking apps, ATM transactions, credit/debit card payments, e-commerce websites, customer accounts, transaction databases.
Actuators (A)	Block suspicious transactions, send alerts to customers, freeze accounts temporarily, notify bank staff, request additional verification (OTP).
Sensors (S)	Transaction records, login information, customer behavior patterns, device information, IP addresses, location data, transaction amount and frequency.

Detailed Explanation About How PEAS Work with example.

1. Performance Measure (P)

This defines how success is measured.

Present in Fraud Detection:

- High fraud detection accuracy.
- Low false positive rate.
- Fast response time.
- Protection of customer funds.
- Customer satisfaction.

Example:

If a hacker tries to transfer Rs. 500,000 from Karachi while the customer normally spends only Rs. 5,000 daily, the system should detect it immediately.

2. Environment (E)

The environment is where the AI system operates.

Present in Fraud Detection:

- Banking applications.
- Digital wallets.

- ATM networks.
- Online shopping websites.
- Financial transaction databases.
- Customers and merchants.

Example:

A customer uses a debit card at a store or online website. The system monitors that environment.

3. Actuators (A)

Actuators are the actions taken by the system.

Present in Fraud Detection:

- Block transaction.
- Send SMS or Email alert.
- Freeze account temporarily.
- Ask for OTP verification.
- Notify fraud investigation team.

Example:

When suspicious activity is detected, the system blocks payment and sends an alert.

4. Sensors (S)

Sensors collect information used by the AI system.

Present in Fraud Detection:

- Transaction amount.
- Transaction time.
- User location.
- Device information.

- Login history.
- IP address.
- Spending behavior.

Example:

The system observes:

- Customer usually logs in from Karachi.
- Suddenly login occurs from another country.
- Large money transfer is requested.

These observations become sensor inputs.

Expected Output

The system should:

- Monitor all banking and payment transactions in real time.
- Identify suspicious or unusual activities.
- Flag potentially fraudulent transactions.
- Generate alerts for customers and bank administrators.
- Block or hold suspicious transactions for verification.
- Reduce financial losses caused by fraud.

Example Output

Transaction ID: 2026001

Amount: Rs. 250,000

Location: Dubai

Risk Score: 95%

Status: Fraud Suspected

Action Taken: Transaction Blocked

Alert Sent to Customer: Yes

Advantages

Security Enhancement

Protects customers from unauthorized transactions.

Real-Time Detection

Identifies fraud immediately.

Financial Loss Reduction

Prevents money theft and banking fraud.

Improved Customer Trust

Customers feel safer using digital banking.

Automated Monitoring

Works 24/7 without human intervention.

Scalability

Can monitor millions of transactions daily.

TASK 1: PEAS Specification

Performance Measures

1. Fraud Detection Accuracy (% of fraudulent transactions correctly identified)
2. False Positive Rate (legitimate transactions incorrectly flagged)
3. Transaction Processing Speed (time taken to analyze a transaction)
4. Financial Loss Prevention (amount of fraud prevented)
5. Customer Satisfaction Score
6. System Availability/Uptime

Environment

The agent operates within digital banking networks, online payment gateways, mobile banking applications, ATMs, e-commerce platforms, and financial transaction databases. External factors include cyberattacks, changing fraud patterns, user behavior, network conditions, and regulatory compliance requirements.

Actuators

1. Transaction Approval Module
2. Transaction Blocking System
3. Fraud Alert Notification Service
4. Customer Verification Request System (OTP, biometric verification)
5. Account Suspension Mechanism
6. Risk Score Generator

Sensors

1. Transaction Records
2. User Login Activity
3. Device Information (IP address, device ID)
4. Geolocation Data
5. Customer Account History
6. Payment Amount and Frequency Data
7. Behavioral Analytics Data
8. Authentication Logs

Task 2: Environment Classification

Fully Observable vs. Partially Observable

Choice: Partially Observable

Justification: The system cannot directly observe a user's true intentions, and some fraud indicators may be hidden or unavailable.

Single-agent vs. multi-agent

Choice: Multi-agent

Justification: The system interacts with customers, banks, merchants, payment gateways, and potential fraudsters.

Deterministic vs. Stochastic

Choice: Stochastic

Justification: Fraudulent behavior is unpredictable and outcomes cannot always be determined with certainty.

Episodic vs. Sequential

Choice: Sequential

Justification: Current decisions depend on previous transactions and account activities.

Static vs. Dynamic

Choice: Dynamic

Justification: Transaction data and fraud patterns continuously change in real time.

Discrete vs. Continuous

Choice: Continuous

Justification: Variables such as transaction amounts, timestamps, and risk scores can take continuous values.

Task 3: Critical Analysis

1. Which dimension poses the greatest technical challenge?

The **Stochastic Environment** poses the greatest challenge because fraudsters constantly develop new techniques to bypass security systems. The AI agent must adapt to uncertain and changing patterns while minimizing false positives and false negatives. Predicting fraudulent behavior is difficult because future actions cannot be known with certainty.

2. Utility Function

A suitable utility function is:

$$U = 0.7(\text{Fraud Detection Accuracy}) - 0.3(\text{Processing Delay})$$

This utility function balances fraud detection accuracy against transaction processing delay.

If the weight of accuracy is doubled:

$$U = 1.4(\text{Fraud Detection Accuracy}) - 0.3(\text{Processing Delay})$$

$$U = 1.4(\text{Fraud Detection Accuracy}) - 0.3(\text{Processing Delay})$$

The system will become more cautious and focus heavily on detecting fraud, even if transaction processing becomes slower. More suspicious transactions may be flagged for verification, reducing fraud risk but increasing customer waiting time.

Task 4: Structured Representation (JSON)

```
{
  "system_name": "AI-Based Fraud Detection System for Digital Payments and Banking Transactions",

  "peas": {
    "performance_measures": [
      "fraud_detection_accuracy",
      "false_positive_rate",
      "transaction_processing_speed",
      "financial_loss_prevention",
      "customer_satisfaction",
      "system_availability"
    ],

    "environment": "Digital banking systems, online payment gateways, mobile banking apps, ATMs, e-commerce platforms, and financial transaction databases with evolving fraud threats.",

    "actuators": [
      "transaction_approval_module",
      "transaction_blocking_system",
      "fraud_alert_notification_service",
      "customer_verification_request_system",
      "account_suspension_mechanism",
      "risk_score_generator"
    ],

    "sensors": [
      "transaction_records",
      "user_login_activity",
```

```
"device_information",
"geolocation_data",
"customer_account_history",
"payment_amount_frequency_data",
"behavioral_analytics_data",
"authentication_logs"
]
},

"environment_classification": {

"observability": {
"choice": "Partially Observable",
"justification": "The system cannot directly observe all fraud-related intentions and hidden
activities."
},

"agents": {
"choice": "Multi-agent",
"justification": "The system interacts with customers, merchants, banks, payment gateways,
and fraudsters."
},

"determinism": {
"choice": "Stochastic",
"justification": "Fraud patterns and attacker behavior are unpredictable."
},

"episodic_sequential": {
"choice": "Sequential",
"justification": "Current transaction decisions depend on previous transaction history."
},

"static_dynamic": {
"choice": "Dynamic",
"justification": "Transaction data and fraud risks change continuously in real time."
},
```

```
"discrete_continuous": {  
  "choice": "Continuous",  
  "justification": "Transaction amounts, risk scores, and time values are continuous."  
}  
,  
  
"utility_function": "U = 0.7*Fraud_Detection_Accuracy - 0.3*Processing_Delay"  
}
```

This version is suitable for a **university AI assignment**, MS Word report, and PDF submission.

Future Enhancements

Artificial Intelligence Improvements

Use Deep Learning models for better detection.

Biometric Authentication

Integrate fingerprint and facial recognition.

Behavioral Analysis

Analyze customer spending habits more accurately.

Blockchain Integration

Increase transaction transparency and security.

Predictive Fraud Detection

Predict fraud before it occurs.

Multi-Factor Authentication

Add OTP, biometrics, and device verification.

Working of the System

Transaction Received



Data Collection



AI Fraud Analysis



Risk Score Generation



Decision Making



Allow or Block Transaction